



SUREN NETWORK
PLAY WITH TECHNOLOGIES

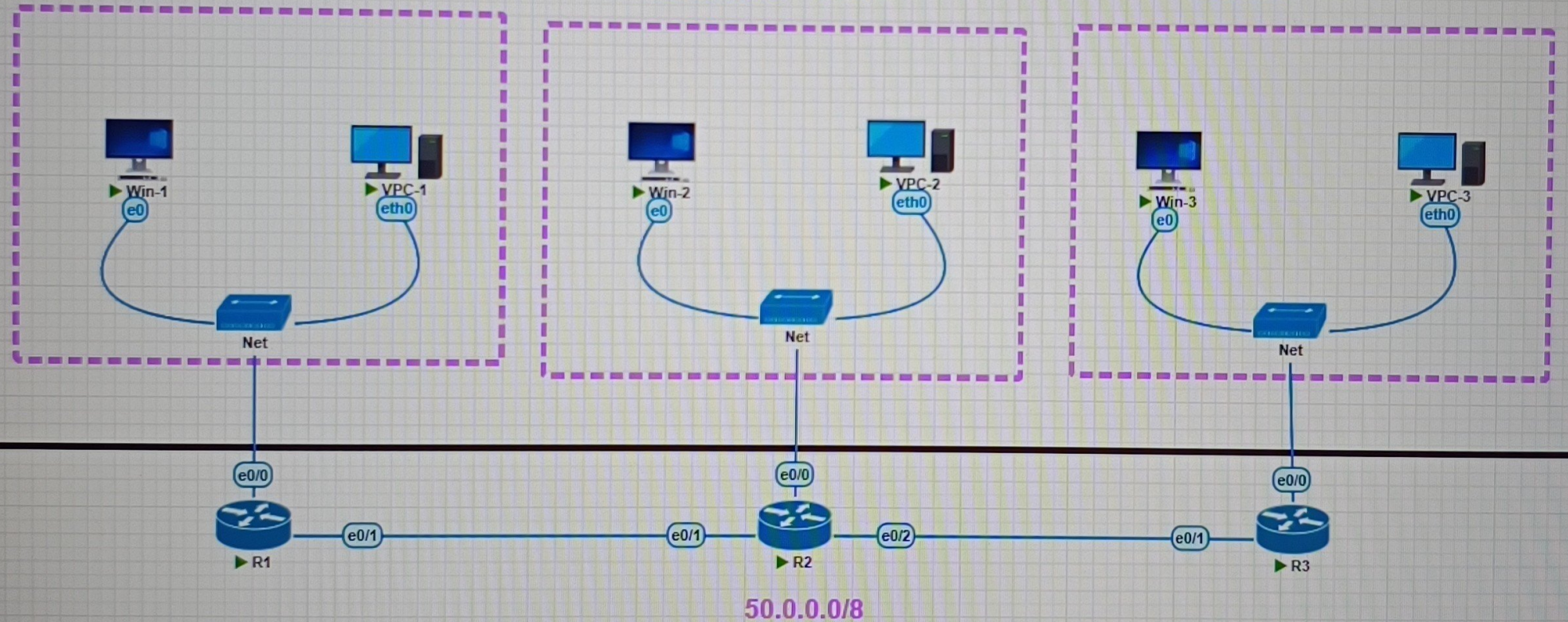
SUBNETTING

HR

50-PC

PRODUCTION 120-PC

SALES 250-PC





R1

```
R1(config)#ex
% Ambiguous command: "ex"
R1(config)#ex
% Ambiguous command: "ex"
R1(config)#exit
R1#
*Sep  9 17:11:34.664: %SYS-5-CONFIG_I: Configured from console by console
R1#sh ip int br
Interface                IP-Address      OK? Method Status      Prot
ocol
Ethernet0/0              50.0.1.129      YES manual  up          up
Ethernet0/1              50.0.1.193      YES manual  up          up
Ethernet0/2              unassigned      YES TFTP   down        down
Ethernet0/3              unassigned      YES TFTP   down        down

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
```

R1

Ethernet0/1	50.0.1.193	YES manual	up	up
Ethernet0/2	unassigned	YES TFTP	down	down
Ethernet0/3	unassigned	YES TFTP	down	down

R1#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

50.0.0.0/8 is variably subnetted, 7 subnets, 5 masks

S	50.0.0.0/24	[1/0]	via 50.0.1.194
S	50.0.1.0/25	[1/0]	via 50.0.1.194
C	50.0.1.128/26	is directly connected, Ethernet0/0	
L	50.0.1.129/32	is directly connected, Ethernet0/0	
C	50.0.1.192/30	is directly connected, Ethernet0/1	
L	50.0.1.193/32	is directly connected, Ethernet0/1	
S	50.0.1.196/30	[1/0]	via 50.0.1.194

R1#

R2#

```
R2(config-if)#ex
```

```
R2(config)#ex
```

```
% Ambiguous command: "ex"
```

```
R2(config)#exit
```

```
R2#
```

```
*Sep  9 17:13:03.743: %SYS-5-CONFIG_I: Configured from console by console
```

```
R2#show ip int br
```

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	50.0.1.1	YES	manual	up	up
Ethernet0/1	50.0.1.194	YES	manual	up	up
Ethernet0/2	50.0.1.197	YES	manual	up	up
Ethernet0/3	unassigned	YES	TFTP	down	down

```
R2#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
```

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
E1 - OSPF external type 1, E2 - OSPF external type 2
```

```
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
```

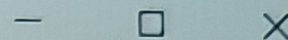
```
ia - IS-IS inter area, * - candidate default, U - per-user static route
```

```
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
```

```
a - application route
```



R2



Ethernet0/1	50.0.1.194	YES manual	up	up
Ethernet0/2	50.0.1.197	YES manual	up	up
Ethernet0/3	unassigned	YES TFTP	down	down

R2#show ip route

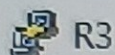
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

50.0.0.0/8 is variably subnetted, 8 subnets, 5 masks

S 50.0.0.0/24 [1/0] via 50.0.1.198
C 50.0.1.0/25 is directly connected, Ethernet0/0
L 50.0.1.1/32 is directly connected, Ethernet0/0
S 50.0.1.128/26 [1/0] via 50.0.1.193
C 50.0.1.192/30 is directly connected, Ethernet0/1
L 50.0.1.194/32 is directly connected, Ethernet0/1
C 50.0.1.196/30 is directly connected, Ethernet0/2
L 50.0.1.197/32 is directly connected, Ethernet0/2

R2#



R3

R3#

R3(config)#

R3(config)#ex

% Ambiguous command: "ex"

R3(config)#exit

R3#

*Sep 9 17:13:31.624: %SYS-5-CONFIG_I: Configured from console by console

R3#sh ip int br

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	50.0.0.1	YES	manual	up	up
Ethernet0/1	50.0.1.198	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

R3#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

```

ocol
Ethernet0/0          50.0.0.1          YES manual up          up
Ethernet0/1          50.0.1.198        YES manual up          up
Ethernet0/2          unassigned        YES TFTP    down          down
Ethernet0/3          unassigned        YES TFTP    down          down

```

R3#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 50.0.1.197 to network 0.0.0.0

```

S* 0.0.0.0/0 [1/0] via 50.0.1.197
50.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C   50.0.0.0/24 is directly connected, Ethernet0/0
L   50.0.0.1/32 is directly connected, Ethernet0/0
C   50.0.1.196/30 is directly connected, Ethernet0/1
L   50.0.1.198/32 is directly connected, Ethernet0/1

```

R3#

R1#

R1(config)#

R1(config)#ip dhcp pool HR

R1(dhcp-config)#network 50.0.1.128 255.255.255.192

R1(dhcp-config)#default-router 50.0.1.129

R1(dhcp-config)#dns-server 8.8.8.8

R1(dhcp-config)#ex

R1(config)#

R1(config)#

R1(config)#ip dhcp pool SALES

R1(dhcp-config)#network 50.0.1.0 255.255.255.128

R1(dhcp-config)#default-router 50.0.1.1

R1(dhcp-config)#dns-server 8.8.8.8

R1(dhcp-config)#ex

R1(config)#

*Sep 9 16:57:12.839: %DHCPD-4-PING_CONFLICT: DHCP address conflict: server pinged 50.0.1.1.

R1(config)#

R1(config)#

R1(config)#ip dhcp pool PRO

R1(dhcp-config)#network 50.0.0.0 255.255.255.0

R1(dhcp-config)#default-router 50.0.0.1

^
% Invalid input detected at '^' marker.

R1(dhcp-config)#default-router 50.0.0.1

R1(dhcp-config)#dns-server 8.8.8.8

R1(dhcp-config)#ex

R1(config)#



R2



```
R2(config)#  
R2(config)#int e0/0  
R2(config-if)#ip helper-address 50.0.1.193  
R2(config-if)#
```

QEMU (Win-1)

C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix	:	:
IP Address	:	50.0.1.130
Subnet Mask	:	255.255.255.192
Default Gateway	:	50.0.1.129

C:\Documents and Settings\user>ping 50.0.0.0

Pinging 50.0.0.0 with 32 bytes of data:

Reply from 50.0.1.198:	bytes=32	time=2ms	TTL=253
Reply from 50.0.1.198:	bytes=32	time=1ms	TTL=253
Reply from 50.0.1.198:	bytes=32	time=1ms	TTL=253
Reply from 50.0.1.198:	bytes=32	time=1ms	TTL=253

Ping statistics for 50.0.0.0:

QEMU (Win-2)

My Documents

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco

C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix	:	:
IP Address	:	50.0.1.2
Subnet Mask	:	255.255.255.128
Default Gateway	:	50.0.1.1

C:\Documents and Settings\user>_